

1. Closed-form LSE approach

N points $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$

The trend line formula:

$$f(x_i) = w_0 + w_1 x_i + w_2 x_i^2 + \dots + w_m x_i^m$$

$$\text{LSE: } \sum [f(x_i) - y_i]^2 = \|Ax - b\|^2$$

$$\begin{pmatrix} 1 & x_1 & x_1^2 & \dots & x_1^m \\ 1 & x_2 & x_2^2 & \dots & x_2^m \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^m \end{pmatrix} \begin{pmatrix} w_0 \\ w_1 \\ \vdots \\ w_m \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

$\downarrow$ $\downarrow$ $\downarrow$
 A x b

- Solve $\arg\min_x \|Ax - b\|^2$

$$\|Ax - b\|^2$$

$$= (Ax - b)^T (Ax - b)$$

$$= (x^T A^T - b^T) (Ax - b)$$

$$= x^T A^T A x - x^T A^T b - b^T A x + b^T b$$

$$= x^T A^T A x - 2 x^T A^T b + b^T b = L(x)$$

$$\frac{dL(x)}{dx} = 2A^T A x - 2A^T b$$

$$\text{To let } \frac{dL(x)}{dx} = 0$$

$$\Rightarrow 2A^T A x - 2A^T b = 0$$

$$\Rightarrow x = (A^T A)^{-1} A^T b$$

2. Steepest descent method

Loss function with L1 norm:

$$E = \sum_{i=0}^n [y_i - (m_k x_i^k + \dots + m_1 x_i + m_0)]^2 + \lambda \sum_{i=0}^k |m_i|$$

We need to find the derivative with respect to $m_0, m_1, \dots, m_k$

$$D_{m_k} = \sum_{i=0}^n 2[y_i - (m_k x_i^k + \dots + m_1 x_i + m_0)] (-x_i^k) + \lambda \frac{d|m_k|}{dm_k}$$

⋮

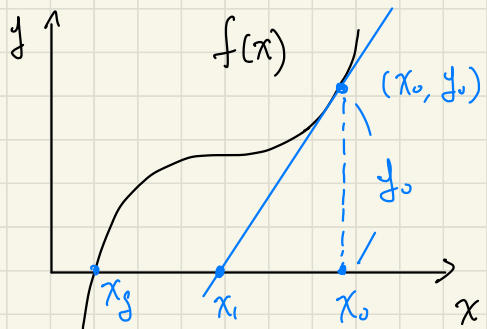
$$D_{m_1} = \sum_{i=0}^n 2[y_i - (m_k x_i^k + \dots + m_1 x_i + m_0)] (-x_i) + \lambda \frac{d|m_1|}{dm_1}$$

$$D_{m_0} = \sum_{i=0}^n 2[y_i - (m_k x_i^k + \dots + m_1 x_i + m_0)] + \lambda \frac{d|m_0|}{dm_0}$$

3. Newton's Method

$$f'(x_0) = \frac{y_0}{x_0 - x_1}$$

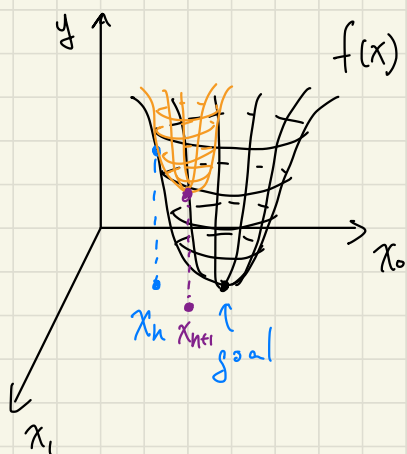
$$x_1 = x_0 - \frac{y_0}{f'(x_0)}$$



- Taylor Expansion

$$f(x) \cong f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2!} f''(x_0)(x-x_0)^2 + \dots$$

$$\cong \sum \frac{f^{(n)}(x_0)}{n!} (x-x_0)^n = g(x)$$



$$f\left(\begin{bmatrix} x_0 \\ x_1 \end{bmatrix}\right) = f$$

$$f(x) \cong f(x_0) + f'(x_0) \overset{x-x_0}{\Delta x} + \frac{1}{2!} f''(x_0) \Delta x^2$$

↑
逼近到二次微分 = g(x)

∴ 在 x_n 這個點

$$g(x) = f(x_n) + f'(x_n) \Delta x + \frac{1}{2!} f''(x_n) \Delta x^2$$

We want to find the x_{n+1} to let $g(x)$ be min

$$g'(x) = f'(x_n) + f''(x_n)(x-x_n) = 0$$

$$\Rightarrow x = \frac{-f'(x_n)}{f''(x_n)} + x_n = x_{n+1}$$

$$x_{n+1} = x_n + \frac{-f'(x_n)}{f''(x_n)} \quad H$$

$$= x_n + -H^{-1}(f(x)) \cdot \nabla f(x_n)$$

Hessian Matrix

$$H(f(x)) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \dots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \dots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

- Use Newton's method to solve $\|Ax - b\|^2$

$$\|Ax - b\|^2 = x^T A^T A x - 2x^T A^T b + b^T b = f(x)$$

$$\nabla f(x) = 2A^T A x - 2A^T b$$

$$H(f(x)) = 2A^T A$$

$$x_{n+1} = x_n - H^{-1}(f(x)) \cdot \nabla f(x_n)$$

$$= x_n - (2A^T A)^{-1} (2A^T A x_n - 2A^T b)$$

$$= x_n - x_n + (A^T A)^{-1} A^T b$$

$$= (A^T A)^{-1} A^T b$$